Tensor Network Solver within a Finite Volume Framework for the 2D Convecting Taylor-Green Vortex

2026 Global Quantum + AI Challenge - Phase 1 Concept Proposal

Problem statement: Airbus, 'Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities'

Public code repository: https://github.com/theaiwillwin/airbus-octonion-tgv-solver

Experimental research output - not a validated engineering deliverable.

1. Problem Framing

Accurately predicting aerodynamic flows requires solving PDEs at HPC scale, and current methods face severe scalability limits: the memory footprint of time-resolved flow solutions grows as O(n^3) with resolution, and expensive wind-tunnel testing remains necessary at demanding operating conditions. Airbus seeks more efficient PDE solvers. We address the prescribed 2D Convecting Taylor-Green Vortex (TGV) because its exact analytical solution enables rigorous, unambiguous measurement of error, runtime, and memory - the three axes on which any claimed advantage must be judged.

Why a quantum-inspired method offers credible advantage here: time-resolved flow solutions are dominated by a modest number of coherent structures, so their space-time tensors admit low-rank tensor-network representations whose storage scales with flow complexity rather than grid resolution. We demonstrate this today, classically, with measured numbers through a 512^2 grid - and the same representation maps directly onto quantum states whose qubit count grows only logarithmically with resolution, giving a concrete, honest bridge to fault-tolerant hardware.

2. Technical Approach

Paradigm: quantum-inspired (Tensor Network within a Finite Volume framework - the second track explicitly permitted by the problem statement), with a mapped pathway to gate-based fault-tolerant hardware. No quantum hardware is used or claimed in Phase 1.

Classical baseline (implemented and verified): a cell-centered finite-volume solver on the periodic [0, 2pi]^2 domain - face-based convective and diffusive fluxes, RK2 time stepping with CFL control, and FFT (spectral) pressure projection enforcing incompressibility. The benchmark truth is the exact analytical convecting TGV solution; no external CFD data is required.

Tensor-network layer (implemented and measured): the full space-time (time, x, y) solution tensor is stored in Tucker-decomposed form (TensorLy) - a tree tensor network. Per-mode ranks can be selected by either standard HOSVD energy truncation or an experimental octonionic control layer: local flow states (velocity, vorticity, velocity gradients) are embedded as 8-component octonions over Fano-plane structure constants, and the exact non-associative associator norm ||(a*b)*c - a*(b*c)|| acts as a relational torsion diagnostic that allocates rank where local flow relationships are complex. We benchmarked the octonion layer against the standard baseline and report the outcome honestly in Section 4 (ablation 2). The compression results below are independent of which rank selector is used.

Quantum pathway: the Tucker/TTN factorization maps onto quantum state preparation with ceil(log2 n) qubits per mode - approximately 46 logical qubits at 64^2 and 57 at 512^2 using our measured ranks (Appendix B). Qubit count grows logarithmically with grid size while classical storage grows linearly: a credible, quantified route to quantum memory advantage under fault-tolerant hardware, stated as a mapping rather than a result.

3. Feasibility and Resource Requirements

The pipeline already runs end-to-end on a standard Windows CPU (NumPy, TensorLy, pandas, matplotlib, pytest; no GPU, cluster, or QPU). All results in this proposal are measured from the public repository above; the largest case (512^2, 4.18 GB dense trajectory) needs ~16 GB RAM and ~20 minutes. Phase 2 needs modest resources: a multi-core CPU node (or small cluster) for 3D and higher-Re extensions, and optional QPU access only for small-scale state-preparation demonstrations.

Honest constraints: (a) the TGV is spectrally narrow and favorable to low-rank storage - our perturbed-flow study measures how the advantage degrades (gracefully) as mode content grows; (b) the reduced-order-model speedup is currently an operation-count estimate, not a wall-clock measurement; (c) current quantum devices cannot execute the required state-preparation circuits - the quantum pathway is a resource-quantified mapping, not a hardware claim.

4. Expected Impact

Phase 1 already delivers measured results rather than promises; a successful Phase 2 PoC converts the measured memory advantage into a time-to-solution advantage and validates it on harder flows. All numbers below regenerate from documented scripts.

Measured result 1 - verified solver: grid-convergence order 1.99 (design 2), divergence ~1e-7, stable across Re = 10-2000 (20x the mandatory Re = 100), 48/48 automated tests passing.

Measured result 2 - tensor-network compression, every row measured:

Grid	Dense	Tucker (guided)	Ratio	Error
64^2	2.1 MB	0.27 MB	7.8x	1.1e-15
128^2	32.8 MB	0.32 MB	102x	1.2e-15
256^2	523 MB	0.52 MB	1010x	1.4e-15
512^2	4.18 GB	0.86 MB	4880x	1.3e-15

Guided ranks saturate at (14, 29, 29) from 128^2 onward - rank tracks flow complexity, not resolution. Our pre-run projection for 512^2 (4887x, from measured rank saturation) matched the subsequent direct measurement (4880x) within 0.15%, validating the projection methodology itself. Dense storage grows O(n^3); compressed storage barely grows.

Measured result 3 - perturbed flows: compression remains substantial on multi-mode perturbed TGV flows (11.1x at 8.0e-8 error at 10% perturbation amplitude; 16.4x at 5.3e-6 at 30%), measuring how gracefully the method degrades as spectral content grows - the direction that matters for industrial flows.

Honest ablation 1 - CFL steering: associator-adaptive time stepping shows ~10% lower L2 error, but a uniform CFL reduction without octonions reproduces it at Re >= 500 at ~60x lower runtime. We do not claim a net steering advantage on this benchmark.

Honest ablation 2 - rank selection: the associator-guided ranks beat the best uniform rank by 1.22-1.73x at matched error, but standard HOSVD singular-value energy truncation - the proper classical adaptive baseline - matches or slightly beats the guidance (6-11% less memory at equal-or-lower error on perturbed flows, ~13x faster selection). We do not claim the associator improves on standard adaptive rank selection; it must demonstrate unique value in Phase 2 or be retired. The compression advantage in result 2 is independent of the rank-selection method and stands either way.

Quantitative Phase 2 targets: (i) measured wall-clock ROM speedup >= 2x via DEIM hyper-reduction; (ii) a decisive associator-vs-POD/adaptive-SVD comparison on spatially heterogeneous flows - with the layer retired if it does not win; (iii) compression >= 100x at <= 1e-6 error on a 3D TGV at 128^3; (iv) small-scale quantum state-preparation demo of a rank-truncated factor on available hardware.

5. Validation Plan

Every claim is validated against the exact analytical TGV solution and a verified classical baseline. Recorded metrics: time-to-solution, memory (dense and compressed), L2 velocity error, kinetic-energy decay, divergence norm, compression ratio, reconstruction error, and associator statistics. The Phase 1 methodology - which Phase 2 will extend - is: (1) verify the solver (grid-convergence order, divergence control); (2) measure, never estimate, wherever possible; (3) run error-matched head-to-head comparisons against the strongest naive baseline; (4) run the control experiment for every claimed advantage and report ablations that fail. Phase 2 success criteria are the four quantitative targets in Section 4, evaluated on the same measured basis, plus independent reproduction from the public repository. Mandatory cases Re = 10 and Re = 100 are complete and extended to Re = 2000 with grid refinement 32^2-512^2; 48 automated tests guard every module.

6. Hybrid / Cross-Domain Integration

The design is hybrid by construction: a classical FV solver produces physics; a tensor-network layer compresses and (in Phase 2) evolves it; the octonionic associator layer supplies geometry-aware control signals; and the TTN factorization is the hand-off point to quantum hardware, where state preparation replaces classical factor storage. Modules are cleanly separated (solver, octonion algebra, diagnostics, compression, benchmarking), so the natural aerodynamic extensions - multi-mode initial conditions, 3D TGV, higher Re - reuse the same algebra and compression pipeline unchanged. Within Airbus workflows, compressed trajectory storage integrates directly with existing HPC post-processing: 4.18 GB -> 0.86 MB per trajectory changes what can be archived, transmitted, and interactively queried from a full unsteady simulation campaign.

7. Team Capability

This submission demonstrates capability by artifact rather than resume: a working, verified solver; a novel octonionic control layer implemented from Fano-plane structure constants (exact associator, no proxy heuristics); measured results through 512^2 with machine-precision reconstruction; control experiments and honest negative results; 48 automated tests; and full public reproducibility. The working style Phase 2 requires - measure, verify, report the failures alongside the wins - is already evidenced throughout this proposal and its repository.

Appendix (supplementary material, 3 pages max)

A. Reproducibility and Rank-Selection Policy

Public repository (all code, tests, configs, results): https://github.com/theaiwillwin/airbus-octonion-tgv-solver

```
python -m pytest                                    # 48/48
python scripts/run_reynolds_sweep.py --config configs/reynolds_sweep.yaml
python scripts/run_advantage_study.py
python scripts/run_scalability_projection.py --re 100 --grids 32 64 128 256
python scripts/run_perturbed_advantage_study.py --re 100 --nx 64
python scripts/run_svd_baseline_study.py --re 100 --nx 64
```

Rank policy: associator_guided_ranks uses a grid-independent cap (min 4, max 32). An earlier internal draft used a grid-dependent cap, which inflated ranks with no error benefit; we corrected the policy, verified by direct decomposition that the saturated ranks (14, 29, 29) reach machine precision at every grid from 128^2 to 512^2, and re-measured all results.

Perturbed-flow detail (Re=100, nx=64): 10% amplitude -> guided ranks (13, 26, 26), 11.1x at 8.0e-8 error, 1.73x vs best uniform rank; 30% -> (11, 22, 22), 16.4x at 5.3e-6, 1.67x. Perturbations are divergence-free multi-mode streamfunction fields.

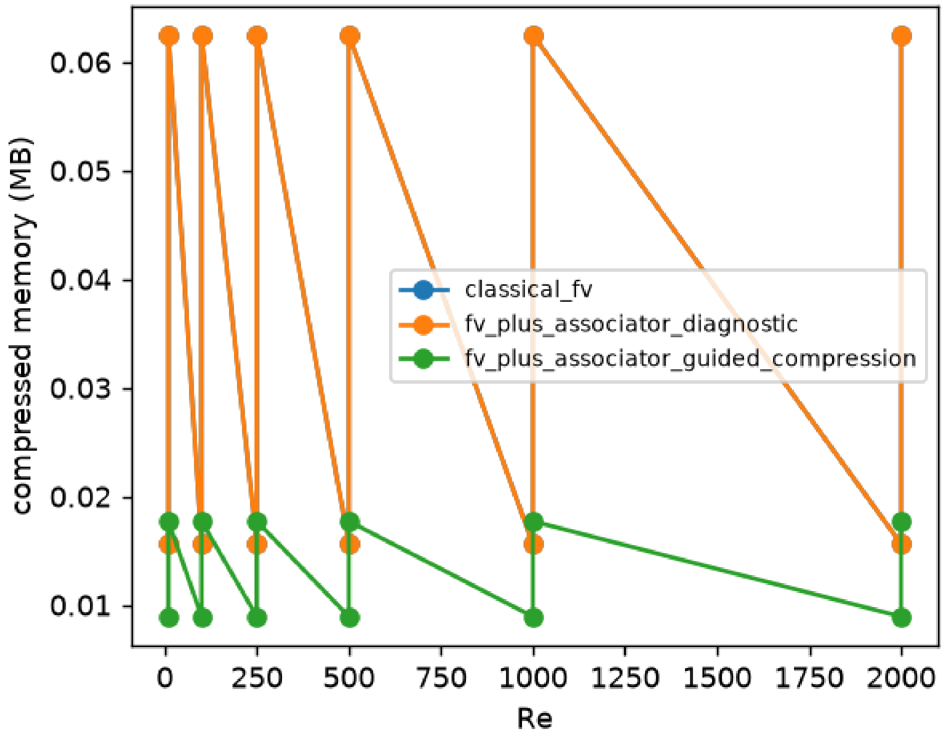
Adaptive-SVD baseline detail (ablation 2): HOSVD energy truncation at matched error selects, e.g., ranks (10, 28, 28) vs guided (13, 26, 26) at 10% perturbation - 0.180 vs 0.192 MB at 3.1e-8 vs 8.0e-8 error - and completes rank selection in ~19 ms vs ~250 ms for the associator diagnostic. Data: results/svd_baseline_*/metrics.csv in the repository.

B. Quantum Hardware Resource Estimates (measured ranks)

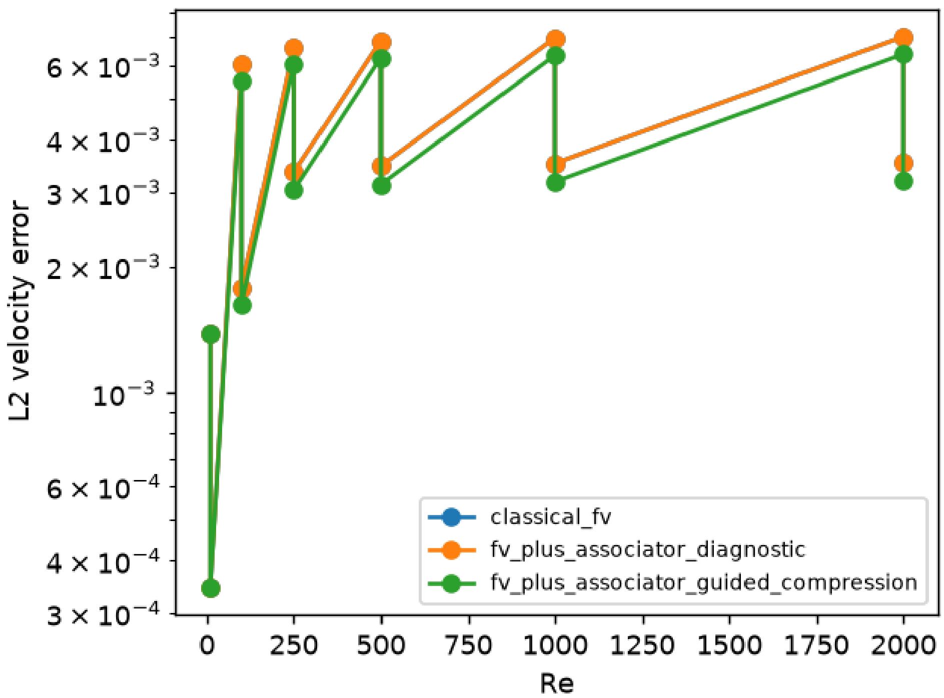
Grid	Ranks (t,x,y)	Logical qubits	Physical qubits*	Basis
64^2	(15, 30, 30)	~46	~46,000	measured
128^2	(14, 29, 29)	~50	~50,000	measured
256^2	(14, 29, 29)	~54	~54,000	measured
512^2	(14, 29, 29)	~57	~57,000	measured

*Surface code, ~1000 physical per logical qubit (d=15). State-preparation circuit depth: O(2x10^4) gates at 64^2, O(1.2x10^5) at 128^2. As nx doubles, spatial-mode qubit count grows by 1 while classical factor storage grows linearly - the logarithmic-vs-linear gap underlying the quantum memory pathway. Current devices (<=133 qubits) cannot execute these circuits; no hardware result is claimed.

C. Key Figures



Memory: dense classical vs compressed storage across the Reynolds sweep.



L2 velocity error vs Reynolds number for all three methods (32^2, 64^2).